

Latent Arithmetic Measures the Origin: Recentering Audio Autoencoders Without Retraining

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Abstract—Audio latents are increasingly judged by arithmetic: interpolate two codes and decode a mix, subtract a stem’s code and decode the residual. We show that the second probe is not a property of the autoencoder alone. Shifting the latent origin by the encoding of silence, $f(0)$, leaves reconstruction and every unit-sum latent combination exactly unchanged for arbitrary nonlinear encoders and decoders, yet moves stem-subtraction scores by decibels. On five fixed checkpoints and a pre-registered held-out corpus of 240 recordings, the shift removes 87–95% of the apparent subtraction advantage of mixing supervision. Applied to autoencoders we did not train, including DAC, EnCodec and Music2Latent, the same shift improves stem subtraction by TBD–TBD dB with no retraining. What mixing supervision genuinely buys is convex mixing, which the shift provably cannot touch: +1.8 and +1.4 dB at two compression rates, replicated across training seeds. Latent arithmetic should be reported against a declared origin.

Index Terms—audio autoencoders, latent arithmetic, representation learning, neural audio codecs, evaluation methodology

I. INTRODUCTION

Mixing is linear on waveforms, $\text{mix}(x_1, x_2, \alpha) = \alpha x_1 + (1-\alpha)x_2$, which makes latent arithmetic an attractive interface to a learned audio representation. For an encoder f and decoder g , the property usually asked for is

$$g(\alpha f(x_1) + (1-\alpha)f(x_2)) \approx \alpha x_1 + (1-\alpha)x_2, \quad (1)$$

and a common companion probe is stem removal by subtraction, $g(f(x_{\text{mix}}) - f(x_{\text{stem}})) \approx x_{\text{res}}$. Where these hold, editing happens in the latent domain without a separation model, and independently generated latents can be composed by addition, one route open to latent-diffusion systems [1], [2]. Reconstruction objectives do not enforce Eq. 1, and Torres et al. [3] induce it in a consistency autoencoder by decoding averaged latents against the true mix.

This paper asks a prior question: *what do these arithmetic scores measure?* The two probes differ in one respect that is easy to overlook. The coefficients in Eq. 1 sum to one; those of subtraction sum to zero. A latent space is defined only up to a choice of origin, and a shift of that origin cancels in the first case but not in the second. Section III makes this exact for arbitrary nonlinear f and g : recentering the latent space at $f(0)$ leaves the autoencoding function and every unit-sum combination pointwise identical, while mapping subtraction to $g(f(x_{\text{mix}}) - f(x_{\text{stem}}) + f(0))$. Any change a subtraction score

shows under this map is therefore a property of the coordinate system, not of the model as a function. Our contributions:

- 1) A reparameterization argument (Section III) showing that subtraction-style latent arithmetic is coordinate-dependent while reconstruction and convex mixing are not, and identifying the model’s own reconstruction of the target as the coordinate-invariant reference.
- 2) The intervention on five fixed checkpoints of a waveform GAN autoencoder (Section V-A), on the test set and on a pre-registered held-out corpus: recentering removes 87–95% of the advantage that mixing supervision appears to have on stem subtraction and lands every checkpoint within half a decibel of its own reconstruction.
- 3) The same shift applied to autoencoders we did not train (Section V-B): DAC [4], EnCodec [5], Music2Latent [6] and the public baselines of [3]. Recentering improves stem subtraction on every one of them without retraining, including the linearity-trained model, whose offset is ten times smaller than its baseline’s but not zero.
- 4) What supervision genuinely buys (Section V-C): convex mixing, which the intervention cannot affect, improves by +1.8 and +1.4 dB at two compression rates, replicates across training seeds, and reduces level error; the widely used decode-versus-decode score is inflated by an adversarial term and by decoder noise (Section V-D).

The practical reading is a two-line recipe: recentre at $f(0)$ before subtracting, whatever the model; add decode-mixing supervision if convex mixing is the goal.

II. RELATED WORK

Neural audio codecs. DAC [4], EnCodec [5] and SoundStream [7] pair convolutional encoders with HiFi-GAN-style decoders [8] and multi-scale discriminators, optimizing reconstruction without latent structure; their residual vector quantization is a further obstacle to interpolation, so we operate on their continuous pre-quantization latents. RAVE [9] learns a variational latent space but does not target mixing equivariance, and disentanglement regularizers [10] encourage generic smoothness rather than a specific identity. Mixing-equivariance follows the symmetry-based view of representation learning [11], though convex mixing is an affine combination, not a group action. **Linear audio representations.** Music2Latent

(M2L) [6] is a consistency autoencoder with partial linearity as a byproduct. Torres et al. [3] make it approximately linear (Lin-CAE) by decoding the average of separately encoded latents against the true mix and by a gain augmentation that enforces homogeneity, a principle related to mixup [12]. Their oracle-separation scores subtract raw latents and compare two decodes, a protocol Section V-B and Section V-D examine. No prior work reports latent arithmetic against a declared origin.

III. THE LATENT ORIGIN IS A FREE COORDINATE

Proposition. Let $f : \mathcal{X} \rightarrow \mathcal{Z}$ and $g : \mathcal{Z} \rightarrow \mathcal{X}$ be arbitrary maps and $c = f(0)$ the encoding of silence. Define the recentered pair

$$f_c(x) = f(x) - c, \quad g_c(z) = g(z + c). \quad (2)$$

Then (i) $g_c(f_c(x)) = g(f(x))$ for every x ; (ii) for any $x_1, \dots, x_n$ and coefficients with $\sum_i a_i = 1$, $g_c(\sum_i a_i f_c(x_i)) = g(\sum_i a_i f(x_i))$; and (iii) for the coefficients $(1, -1)$,

$$g_c(f_c(x) - f_c(y)) = g(f(x) - f(y) + f(0)). \quad (3)$$

Proof. $f_c(x) + c = f(x)$ gives (i). For (ii), $\sum_i a_i f_c(x_i) = \sum_i a_i f(x_i) - c \sum_i a_i = \sum_i a_i f(x_i) - c$, and g_c adds c back. For (iii) the coefficients sum to zero, so c cancels in the latent difference and g_c adds it inside g . $\square$

Three consequences shape the paper. First, (f_c, g_c) is the same autoencoder as (f, g) as far as reconstruction and Eq. 1 are concerned: by (i) and (ii) those quantities are pointwise identical, for arbitrary nonlinear networks, with no retraining and no approximation. Second, by (iii) subtraction is not: recentering shifts it by one encode of silence, and Section V-A measures a several-decibel shift. A model ranking obtained from raw subtraction can therefore be altered by a change of coordinates that provably leaves reconstruction and mixing untouched, so such a ranking is not by itself a statement about the representation. Third, when f is affine, $f(x) = Ax + b$, we have $f(x) - f(y) + f(0) = f(x - y)$ and Eq. 3 decodes exactly $g(f(x_{\text{res}}))$, the model’s own reconstruction of the residual, whatever the decoder does. That encode–decode reconstruction is the reference we report against; the shortfall from it is the *linearity tax*. Real encoders depart from affine, so for them any improvement from recentering is an empirical finding; homogeneity $f(ax) = af(x)$, if trained for, drives $f(0)$ toward zero but need not remove the effect. The proposition concerns re-expressing a *frozen* model: it does not make training origin-invariant, and a latent ℓ_2 penalty in particular can move $f(0)$.

IV. EXPERIMENTAL SETUP

A. Models

Our checkpoints. The primary model is a waveform autoencoder (DAC-style 1-D convolutional encoder, HiFi-GAN decoder [8], multi-scale-STFT and multi-period discriminators) at 44.1 kHz on 1-second mono chunks, trained on a

~ 950 -hour union of FMA-large, MAESTRO and MUSDB18-HQ [13].¹ Two compression rates are used: $7.66 \times$ ($d=128$) and $15.3 \times$ ($d=64$). Mixing supervision is the explicit-loss form of the additivity principle of [3]: pairs (x_i, x_j) from a fixed-point-free permutation, $\alpha \sim U[0, 1]$, $\bar{x} = \alpha x_i + (1 - \alpha)x_j$, $\bar{z} = \alpha f(x_i) + (1 - \alpha)f(x_j)$, and

$$\mathcal{L}_{\text{dec}} = \mathcal{L}_{\text{recon}}(g(\bar{z}), \bar{x}) + \|g(\bar{z}) - \bar{x}\|_1, \quad (4)$$

where $\mathcal{L}_{\text{recon}}$ is the multi-resolution STFT and mel loss also used for reconstruction; the mixed path carries the extra unweighted waveform ℓ_1 term, the reconstruction path does not. Gradients flow through g and into f . The total objective is $\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{GAN}} + \lambda \mathcal{L}_{\text{dec}} + \lambda_{\ell_2} \|z\|^2$ with $\lambda=0.5$ and $\lambda_{\ell_2}=10^{-3}$; a variant also applies the discriminator to the mixed path (“+disc”). At $7.66 \times$ the mixing arms are 25k-step fine-tunes of a converged reconstruction-only checkpoint against a continued-training control; at $15.3 \times$ the combined $\mathcal{L}_{\text{dec}}$ +disc recipe and its control are trained from scratch for 250k steps under matched configurations. The five resulting checkpoints, plus the reconstruction-only ancestor, are frozen for everything below.

Third-party autoencoders. We apply the intervention, unchanged, to public models we had no hand in: DAC at 44.1 kHz [4] and EnCodec at 24 kHz [5], using their continuous encoder outputs with the quantizer bypassed on every path; the published Music2Latent checkpoint [6]; and the three public models of [3]: their M2L retrained on their corpus, Lin-CAE, and Lin-CAE-2. None of these received any change of weights. EnCodec is resampled to and from 24 kHz, which band-limits every route equally.

B. Metrics

SI-SDR [14] is the waveform score throughout. Stem removal is *oracle*: the removed stem is available and encoded, so the probe tests the representation’s arithmetic, not blind separation. Raw subtraction decodes $f(x_{\text{mix}}) - f(x_{\text{stem}})$; the recentered form adds $f(0)$ as in Eq. 3. Both are scored against the true residual, and against each model’s own reconstruction of it (the reference of Section III); the difference is the linearity tax. For consistency decoders, which draw fresh noise per call and never align in phase with a raw waveform, we report the decode-versus-decode form with shared noise, as [3] do against autoencoded ground truth. For convex mixing at $\alpha=0.5$ we report $\text{SI-SDR}_{\text{lin}}^{\text{gt}} = \text{SISDR}(g(\bar{z}), \bar{x})$ (against the true mix), $\text{SI-SDR}_{\text{lin}} = \text{SISDR}(g(\bar{z}), g(f(\bar{x})))$ (decode versus decode), reconstruction $\text{SI-SDR}_{\text{rec}}$, and MixRate, the ratio of the mixed-path loss of Eq. 4 for $g(\bar{z})$ to that for $g(f(\bar{x}))$ (< 1 beats the encode-then-decode path). FAD uses music-finetuned LAION-CLAP embeddings of 10s windows at 48 kHz and is not comparable with VGGish FAD.

C. Protocol and what the intervals cover

Subtraction uses 49 of the 50 MUSDB18 test tracks (one is excluded because its stems sum to the mixture at only

¹Code, configurations, weights, every evaluation record and audio examples: <https://github.com/nurdauletakhanov/musicgen>

Model	drums	bass	vocals	other	all	gap↓
GAN, 7.66×, no mix	+4.2	+0.4	+5.2	+4.0	+3.4	5.2
+ $f(0)$	+8.8	+6.8	+9.6	+8.0	+8.3	0.4
GAN, 7.66×, + $\mathcal{L}_{\text{dec}}$	+5.7	+4.9	+7.3	+5.2	+5.8	2.9
+ $f(0)$	+8.5	+6.9	+9.4	+8.3	+8.3	0.5
GAN, 7.66×, + $\mathcal{L}_{\text{dec}}$ +disc	+6.2	+5.4	+7.5	+5.2	+6.1	2.6
+ $f(0)$	+8.5	+7.0	+9.3	+8.1	+8.2	0.4
GAN, 15.3×, no mix	+1.2	-1.6	+3.0	+2.2	+1.2	6.6
+ $f(0)$	+7.3	+5.8	+8.8	+7.2	+7.3	0.5
GAN, 15.3×, + $\mathcal{L}_{\text{dec}}$ +disc	+5.4	+4.7	+6.9	+4.0	+5.2	2.4
+ $f(0)$	+7.3	+6.1	+8.5	+7.2	+7.3	0.4
M2L, +mix	-9.9	-12.2	-9.9	-12.4	-11.1	8.3
+ $f(0)$	-5.4	-8.0	-5.3	-7.5	-6.5	3.8

TABLE I

STEM REMOVAL BY LATENT SUBTRACTION (SI-SDR, dB; MUSDB18 TEST, 5,880 UNITS FROM 49 TRACKS). “+ $f(0)$ ” ROWS READ THE *same* CHECKPOINT IN RECENTERED COORDINATES. “GAP” IS THE LINEARITY TAX AGAINST EACH MODEL’S OWN ENCODE-DECODE RECONSTRUCTION OF THE RESIDUAL (LOWER IS BETTER). M2L DENOTES MUSIC2LATENT AFTER MIXED-LATENT FINE-TUNING.

18.5 dB), 30 windows per track, four stems: 5,880 (track, stem, chunk) units. Window selection is a pure function of the seed and the sorted track list, so every model, ours or not, sees identical units and every contrast is paired. Held-out evaluation uses MoisesDB [15], 240 recordings absent from the training data and from checkpoint selection of every model evaluated on it, under a protocol registered before the audio was downloaded, with frozen weights and verified hashes; a unit is scored where the removed stem is active (24,648 of 28,800). MUSDB18 is training data for DAC and MoisesDB is training data for the Lin-CAE family, so each third-party model is reported on the corpus it has not seen. Confidence intervals come from a paired cluster bootstrap over recordings (10^4 resamples); mixing pairs join two chunks, so those intervals weight a pair by the multiplicity of each member. Intervals are a conditional diagnostic: they cover evaluation variance for these checkpoints and cover neither training-seed variance nor checkpoint selection. That selection needs disclosure: our released weights are the lowest-validation-loss checkpoint and the validation loader read the test split. For every 7.66× run this coincides with the final step; both 15.3× runs were selected at step 240k of 250k. The origin intervention compares a checkpoint with itself, so it is unaffected; between-model contrasts are why the held-out corpus exists.

V. RESULTS

A. Recentring removes the subtraction advantage

In raw coordinates the supervision looks decisive. It raises subtraction from +3.4 to +5.8 dB at 7.66× and from +1.2 to +5.2 dB at 15.3× (Table I).

Recentring removes most of it. Adding $f(0)$ gains +2.0 to +6.0 dB for every configuration and leaves each within 0.4–0.5 dB of its own reconstruction reference; the models that

Treatment vs. its control	raw	recent.	95% CI	tracks
<i>MUSDB18 test (49 tracks)</i>				
$\mathcal{L}_{\text{dec}}$, 7.66×	+2.33	-0.03	[-0.15, +0.10]	49/49
$\mathcal{L}_{\text{dec}}$ +disc, 7.66×	+2.64	-0.07	[-0.19, +0.07]	49/49
$\mathcal{L}_{\text{dec}}$ +disc, 15.3×	+4.02	+0.01	[-0.19, +0.21]	49/49
<i>MoisesDB, held out (240 tracks)</i>				
$\mathcal{L}_{\text{dec}}$, 7.66×	+2.27	+0.29	[+0.20, +0.39]	229/240
$\mathcal{L}_{\text{dec}}$ +disc, 7.66×	+2.47	+0.12	[+0.03, +0.22]	231/240
$\mathcal{L}_{\text{dec}}$ +disc, 15.3×	+4.50	+0.42	[+0.29, +0.55]	236/240

TABLE II

THE INTERVENTION ON FIXED CHECKPOINTS. ADVANTAGE IN STEM-SUBTRACTION SI-SDR (dB) OF EACH MIXING-TRAINED MODEL OVER ITS MATCHED NO-MIXING CONTROL, RAW AND RECENTERED, WITH THE 95% RECORDING-BOOTSTRAP INTERVAL OF THE RECENTERED ADVANTAGE AND THE NUMBER OF TRACKS ON WHICH THE GAP MAGNITUDE SHRINKS. NO RETRAINING, SAME UNITS, SAME AUDIO.

gain most looked worst. The between-model advantage falls to -0.03, -0.07 and +0.01 dB, all three intervals containing zero (Table II), and the reduction holds on 49 of 49 tracks in every pair. The no-mixing models carry about three times the offset ($\|f(0)\|$ 18.1 versus 6.6 at 7.66×, 14.8 versus 5.4 at 15.3×): much of what mixing supervision did to raw subtraction was to shrink an offset that a change of coordinates removes for free.

Held-out replication, and a residue. On MoisesDB, raw subtraction again favours the mixing models by +2.3 to +4.5 dB and recentring removes 87–95% of that, shrinking the gap on 229, 231 and 236 of the 240 recordings (Table II); on the rest it overshoots and reverses the sign. The residue is resolvable here, +0.29, +0.12 and +0.42 dB with intervals excluding zero, where the 49-track set could not. Recentring lifts each held-out score to within 0.5–1.0 dB of its reference.

Baselines without latent arithmetic. The unchanged mixture scores +17.1 dB against the residual, above every model’s reconstruction reference, because the removed stem is a small share of the energy: the metric certifies arithmetic relative to that reference, not removal. Recentred latent subtraction beats decode-then-subtract, $g(f(x_{\text{mix}})) - g(f(x_{\text{stem}}))$, by +0.7 to +0.9 dB; raw subtraction loses to it everywhere.

B. Recentring models we did not train

The proposition is model-agnostic, so we apply the shift to autoencoders with no mixing supervision and no relation to our training runs (Table III). Our reconstruction-only ancestor gains +5.0 dB [+4.5, +5.6] and lands 0.3 dB from its reference. The codecs carry large offsets ($\|f(0)\|$ per element 2.9 for DAC, 3.7 for EnCodec, against 0.09–0.28 for our models) and gain accordingly: on the held-out corpus DAC gains TBD dB and EnCodec TBD dB, and on MUSDB18 test +5.4 [+4.6, +6.3] and +14.5 [+13.9, +15.1] dB. Both remain short of their references and, unlike our models, below decode-then-subtract in the waveform; recentring removes the offset, not the nonlinearity. For the consistency decoders we read the phase-cancelled score. Music2Latent moves from

Model	raw	$+f(0)$	Δ	ref.
GAN AE, recon. only (ours)	+3.2	+8.1	+5.0	+8.5
Music2Latent [6] [†]	-3.2	+0.3	+3.5	-2.7
M2L retrained [3] [†]	-2.4	+1.3	+3.7	-0.2
Lin-CAE [3] [†]	+11.2	+18.3	+7.0	-0.3
Lin-CAE-2 [3] [†]	+10.9	+18.5	+7.6	-0.3

TABLE III

RECENTERING ON AUTOENCODERS TRAINED WITHOUT ANY LINEARITY OBJECTIVE, AND ON THE PUBLIC MODELS OF [3].

LATENT STEM SUBTRACTION (SI-SDR, dB) IN RAW AND RECENTERED COORDINATES, THE PAIRED CHANGE, AND THE REFERENCE: EACH MODEL’S OWN RECONSTRUCTION OF THE RESIDUAL, SCORED AGAINST THE TRUE RESIDUAL. *MOISESDB HELD-OUT CORPUS (MUSDB IS DAC TRAINING DATA); ALL OTHERS MUSDB18 TEST. [†]CONSISTENCY DECODERS: RAW AND RECENTERED SCORES ARE DECODE-VERSUS-DECODE WITH SHARED NOISE, AS IN [3]; THEIR NEAR-ZERO REFERENCE SHOWS THE PHASE INCOHERENCE THAT MAKES THE WAVEFORM SCORE UNINFORMATIVE.

−3.2 to +0.3 dB (+3.5 [+1.8, +5.2]) and its mixed-latent fine-tune from +0.4 to +8.9 dB (+8.5 [+6.9, +10.1]). The baseline retrained by [3] gains +3.7 dB [+2.1, +5.5]; Lin-CAE and Lin-CAE-2, whose encoding of silence has per-element RMS 0.16 against 1.5 for that baseline, still gain +7.0 [+5.9, +8.3] and +7.6 [+6.4, +9.0] dB and reach 18 dB. The linearity training therefore buys subtraction quality that the coordinate change does not, and the coordinate change buys quality that the training does not: both models improve, and the gap between them does not close.

C. Convex mixing tells a different story

By part (ii) of the proposition, recentering cannot change any quantity in Eq. 1. The mixing probe therefore reports on the autoencoder itself, and there the supervision does something. On the held-out corpus $\mathcal{L}_{\text{dec}}$ improves ground-truth mixing SI-SDR by +1.77 dB [+1.73, +1.81] at $7.66\times$ and the combined recipe by +1.39 dB [+1.33, +1.45] at $15.3\times$, at reconstruction changes of +0.04 and −0.07 dB. On the test set $\mathcal{L}_{\text{dec}}$ raises SI-SDR_{lin}^{gt} from 8.0 to 10.2 dB and is the only configuration with MixRate below one (0.97 against 1.20), with reconstruction SI-SDR spanning 11.2–11.4 dB and FAD 0.044–0.046 across all fine-tunes. An independent re-run of the control and of $\mathcal{L}_{\text{dec}}$ from the same ancestor gives 8.1 and 10.3 dB, so the 2.2 dB test-set gain replicates across training seeds. From scratch at $15.3\times$ the combined recipe improves 6.8 to 8.6 dB and MixRate 1.19 to 1.05 at equal reconstruction, while FAD rises from 0.052 to 0.061. Across a five-point α sweep the gain holds at every coefficient and the SI-SDR range shrinks from 3.2 to 1.9 dB. **Level, not only shape:** the gain SI-SDR discards shows the decoded mix 1.6 dB too quiet without the loss and 0.9 dB with it. Decoder-only adaptation with a frozen encoder retains 9.5 of the 10.2 dB, and encoder-side supervision $\mathcal{L}_{\text{enc}} = \|f(\bar{x}) - \bar{z}\|^2$ lowers the latent error but reaches only 8.4–8.7 dB: the property lives in the decoder’s treatment of interpolated latents. **The mixed-path discriminator buys the metric, not the property.** At

$7.66\times$ it raises decode-versus-decode SI-SDR_{lin} from 12.1 to 16.0 dB while ground-truth mixing falls from 10.2 to 9.9 dB and MixRate rises to 1.05; held out, agreement rises by +3.98 dB while fidelity falls by 0.49 dB [−0.51, −0.48]. The two measurements rank these checkpoints differently and are not interchangeable evidence.

D. Decode-versus-decode scores are protocol-sensitive

For stochastic decoders the decode-versus-decode metric is confounded a second way. A consistency-model decoder [6] draws fresh noise per call: decoding the *same* latent twice is deterministic under shared noise but scores −1.8 dB under independent noise, with no latent arithmetic at all. On identical pairs the same two checkpoints score −9.1 and −7.3 dB with independent noise and +3.6 and +5.9 dB when they share it: a 13 dB swing driven by the protocol rather than the model. Which protocol prior work used is not stated in enough detail for us to determine. We share decode noise everywhere and, for deterministic decoders, score against the true waveform.

VI. CONCLUSION

Latent stem subtraction is measured in coordinates that the model never fixes. Recentering at the encoded silence leaves the autoencoding function and every unit-sum combination pointwise unchanged, for arbitrary nonlinear encoders and decoders, yet it moves subtraction scores by several decibels: it removes 87–95% of the advantage mixing supervision appeared to have on a pre-registered 240-track corpus, and it improves stem subtraction on DAC, EnCodec, Music2Latent and our own reconstruction-only model with no retraining. What the supervision buys is convex equivariance, which the intervention cannot affect and which improves at both rates and across seeds. Two practices follow: report latent arithmetic against a declared origin and against the model’s own reconstruction of the target, and recentre before subtracting, whatever the model. **Limitations.** Recovery is exact only for an affine encoder; for the networks here the gain is empirical and leaves a residue. Comparisons use single training runs except where stated; intervals cover recording-level evaluation variance for fixed checkpoints, not variation across training runs. Selection used the test split; audio is mono, subtraction is oracle, and no listening test was run.

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